

Exercise J01

Q1. Let $A + 2B = \begin{vmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{vmatrix}$ and $2A - B = \begin{vmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{vmatrix}$

then $\text{Tr}(A) - \text{Tr}(B)$ has the value equal to

- (A) 0 (B) 1 (C) 2 (D) None

Ans. C

Sol. $t_r(A) + 2 t_r(B) = -1$ (from the given matrix) and $2t_r(A) - t_r(B) = 3$ (from the given matrix)

Let $t_r(A) = x$ and $t_r(B) = y$

$$x + 2y = -1$$

$$2x - y = 3$$

solving $x = 1$ and $y = -1$

Hence $t_r(A) - t_r(B) = x - y = 2$

Q2. If A and B are two matrices such that , $AB = B$ and $BA = A$, then $A^2 + B^2 =$

- (A) 2 AB (B) 2BA (C) A+B (D) AB

Ans. C

Sol. -

Q3. Let $f(x) = x^2 - 5x + 6$ and $A = \begin{vmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{vmatrix}$, then $f(A)$ is equal to :

(A) $\begin{vmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 4 \end{vmatrix}$ (B) $\begin{vmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 3 \end{vmatrix}$ (C) $\begin{vmatrix} 0 & -1 & -3 \\ 0 & -1 & -10 \\ -5 & 4 & 3 \end{vmatrix}$ (D) $\begin{vmatrix} 0 & -1 & -3 \\ 0 & -1 & -10 \\ -5 & 4 & 0 \end{vmatrix}$

Ans. A

Sol. -

Q4. If $A = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$, then $A^2 = B^2 = C^2 =$

- (A) I^2 (B) I (C) $-I$ (D) $2I$

Ans. C

Sol. -

Q5. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 2 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -5 & 7 & 1 \\ 1 & -5 & 7 \\ 7 & 1 & -5 \end{bmatrix}$ then $AB =$

- (A) I_3 (B) $2 I_3$ (C) $4 I_3$ (D) $18 I_3$

Ans. D

Sol. -

Q6. If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$, then

- (A) $\alpha = 2ab, \beta = a^2 + b^2$ (B) $\alpha = a^2 + b^2, \beta = ab$ (C) $\alpha = a^2 + b^2, \beta = 2ab$
 (D) $\alpha = a^2 + b^2, \beta = a^2 - b^2$

Ans. C

Sol. -

Q7. If A and B are square matrices of size $n \times n$ such that $A^2 - B^2 = (A - B)(A + B)$, then which of the following will be always true –

- (A) $AB = BA$ (B) Either of A or B is a zero matrix
 (C) Either of A or B is an identity matrix (D) $A = B$

Ans. A

Sol. -

Q8. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, $a, b \in \mathbb{N}$. Then –

- (A) there exist more than one but finite number of B's such that $AB = BA$
- (B) there exist exactly one B such that $AB = BA$
- (C) there exist infinitely many B's such that $AB = BA$
- (D) there cannot exist any B such that $AB = BA$

Ans. C

Sol. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

$$B = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \quad a, b \in \mathbb{N}$$

$$AB = \begin{pmatrix} a & 2b \\ 3a & 4b \end{pmatrix}$$

$$BA = \begin{pmatrix} a & 2b \\ 3b & 4b \end{pmatrix}$$

Natural number for which $a = 6$

so Infinite matrix B possible

Q9. Let $A = \begin{bmatrix} 5 & 5\alpha & \alpha \\ 0 & \alpha & 5\alpha \\ 0 & 0 & 5 \end{bmatrix}$ If $|A^2| = 25$, then $|\alpha|$ equals-

- (A) 5^2 (B) 1 (C) $\frac{1}{5}$ (D) 5

Ans. C

Sol. $|A^2| = 25$

$$|A|^2 = 25$$

$$|A| = 5$$

$$\begin{bmatrix} 5 & 5\alpha & \alpha \\ 0 & \alpha & 5\alpha \\ 0 & 0 & 5 \end{bmatrix} = 5$$

$$25\alpha = 5$$

$$\alpha = \frac{1}{5}$$

Q10. If $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and $|A^3| = 125$, then $\alpha =$

- (A) ± 3 (B) ± 2 (C) ± 5 (D) 0

Ans. A

Sol. -

Q11. A and B are two given matrices such that the order of A is 3×4 , if $A'B$ and BA' are both defined then

- (A) order of B' is 3×4 (B) order of $B'A$ is 4×4 (C) order of $B'A$ is 3×3
(D) $B'A$ is undefined

Ans. B

Sol. $A = 3 \times 4$; $A' = 4 \times 3$

As $A'B$ is defined $\Rightarrow$ let order of $B = 3 \times n$

now $BA' = (3 \times n) \times (4 \times 3) \Rightarrow n = 4$

$\therefore$ order of B is 3×4

$\therefore$ order of $B' = 4 \times 3$

order of $B'A = (4 \times 3) \times (3 \times 4) = 4 \times 4$ Ans.

Q12. If $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, A^{-1} is given by

- (A) $-A$ (B) A^T (C) $-A^T$ (D) A

Ans. B

Sol. For Adj interchange the diagonal elements and change the sign of off diagonal elements.

We have $A^{-1} = \frac{1}{\cos^2 \theta + \sin^2 \theta} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = A^T \Rightarrow A$ is orthogonal matrix

Q13. For a given matrix $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ which of the following statement holds good?

- (A) $A = A^{-1} \forall \theta \in \mathbb{R}$ (B) A is symmetric, for $\theta = (2n + 1) \frac{\pi}{2}$, $n \in \mathbb{I}$
(C) A is an orthogonal matrix for $\theta \in \mathbb{R}$ (D) A is a skew symmetric, for $\theta = n\pi$; $n \in \mathbb{I}$

Ans. C

Sol. Obv. A is orthogonal as $a_{11}^2 + a_{12}^2 = 1 = a_{21}^2 + a_{22}^2$

for skew symmetric matrix $a_{ii} = 0 \Rightarrow \theta = (2n + 1) \frac{\pi}{2}$

for symmetric matrix, $A = A^T \Rightarrow \sin \theta = 0 \Rightarrow \theta = n\pi$

Also $\text{adj } A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ and $|A| = 1$ hence $A = A^{-1}$ is possible if $\sin \theta = 0$

Q14. If A is an orthogonal matrix, then A^{-1} equals :

(A) A (B) A' (C) A^2 (D) None of these

Ans. B

Sol. -

Q15. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying the equation $AA^T = 9I$, where I is 3×3 identity matrix, then the ordered pair (a, b) is equal to :

(A) (2, -1) (B) (-2, 1) (C) (2, 1) (D) (-2, -1)

Ans. D

Sol. All rows of A represents components of 3 vectors which are mutually perpendicular and magnitude is 3

a must be 2 or -2

b must be 1 or -1

$$\begin{array}{ll} 4 - 2 + 2b = 0 & 2 + 2 + 2a = 0 \\ b = -1 & a = -2 \end{array}$$

Q16. If A is a square matrix of order n then, $\text{adj}(\text{adj } A)$ is equal to :

(A) $|A|^n A$ (B) $|A|^{n-1} A$ (C) $|A|^{n-2} A$ (D) $|A|^{n-3} A$

Ans. C

Sol. -

Q17. If $A = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$ and $A \text{adj } A = k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ the value of 'k' is :

(A) $\sin x \cos x$ (B) 1 (C) 2 (D) 3

Ans. B

Sol. -

Q18. If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and $A \text{adj } A = A A^T$, then $5a + b$ is equal to :

(A) 4 (B) 13 (C) -1 (D) 5

Ans. D

Sol. $A \operatorname{adj} A = A A^T$
 $\operatorname{adj} A = A^T$

$$\begin{bmatrix} 2 & b \\ -3 & 5a \end{bmatrix} = \begin{bmatrix} 5a & 3 \\ -b & 2 \end{bmatrix}$$
$$2 = 5a, b = 3$$

Q19. If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$, then $\operatorname{adj} (3A^2 + 12A)$ is equal to

(A) $\begin{bmatrix} 51 & 54 \\ 63 & 72 \end{bmatrix}$ (B) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$ (C) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$ (D) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$

Ans. D

Sol. $(3A^2 + 12A) = 3A(A + 4I)$

$$\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$$

$$\operatorname{adj} (3A^2 + 12A) = \begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$$

Q20. Let $A = \begin{bmatrix} 2 & 3 \\ -1 & 5 \end{bmatrix}$. If $A^{-1} = xA + yI_2$, (I_2 is Identity matrix of 2×2) then the value of x and y can be

(A) $x = -\frac{1}{13}$, $y = \frac{7}{13}$ (B) $x = \frac{1}{13}$, $y = \frac{7}{13}$ (C) $x = \frac{1}{13}$, $y = \frac{7}{13}$
(D) None of these

Ans. A

Sol. -

Q21. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$, $I =$ and $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A^{-1} = \frac{1}{6}(A^2 + cA + dI)$, then the value of c and d are

(A) $-6, -11$ (B) $6, 11$ (C) $-6, 11$ (D) $6, -11$

Ans. C

Sol. -

Q22. A is an involutory matrix given by $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ then the inverse of $\frac{A}{2}$ will be

(A) $2A$ (B) $\frac{A^{-1}}{2}$ (C) $\frac{A}{2}$ (D) A^2

Ans. A

Sol. A is involutory $\Rightarrow A^2 = I \Rightarrow A = A^{-1}$

$$\text{now } A^2 = \left(\frac{A}{2}\right) (2A) = I \Rightarrow 2A = \left(\frac{A}{2}\right)^{-1}$$

Q23. Which of the following statements is incorrect for a square matrix A. ($|A| \neq 0$)

- (A) If A is a diagonal matrix, A^{-1} will also be a diagonal matrix
- (B) If A is a symmetric matrix, A^{-1} will also be a symmetric matrix
- (C) If $A^{-1} = A \Rightarrow A$ is an idempotent matrix
- (D) If $A^{-1} = A \Rightarrow A$ is an involutory matrix

Ans. C

Sol. $A^2 = I \rightarrow$ Involutary Matrix

$A^2 = A \rightarrow$ Idempotent Matrix

Q24. Let the matrix A and B be defined as $A = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ 7 & 3 \end{bmatrix}$ then the value of $\text{Det.}(2A^9B^{-1})$, is

- (A) 2 (B) 1 (C) -1 (D) -2

Ans. D

Sol. $\text{Det.}(2A^9B^{-1}) = \frac{2^2(\text{Det. } A)^9}{\text{Det. } B} = \frac{2^2(-1)^9}{2} = -2$ Ans.

Q25. If A and B are invertible matrices, which one of the following statements is not correct

- (A) $\text{Adj. } A = |A| A^{-1}$ (B) $\det(A^{-1}) = |\det(A)|^{-1}$ (C) $(A+B)^{-1} = B^{-1} + A^{-1}$
- (D) $(AB)^{-1} = B^{-1}A^{-1}$

Ans. C

Sol. $AA^{-1} = I \Rightarrow |AA^{-1}| = |I| = 1$

Hence $|A| |A^{-1}| = 1 \Rightarrow \frac{1}{|A|} |A^{-1}| \Rightarrow (B) \text{ is correct}$

Q26. If $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} A \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then the matrix A is equal to :

- (A) $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Ans. C

Sol. -

Q27. If $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, then the value of $|A' A^{-1}|$ is :

- (A) $\cos 4x$ (B) $\sec^2 x$ (C) $-\cos 4x$ (D) 1

Ans. D

Sol. -

Q28. Which of the following is incorrect ?

- (A) $\text{adj}(\text{adj } A) = A$ (B) $(A^T)^T = A$ (C) $(A^{-1})^T = (A^T)^{-1}$
 (D) $(A - I)(I + A) = 0 \Leftrightarrow A^2 = I$

Ans. A

Sol. -

Q29. Let $A = \begin{pmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$. The only correct statement about the matrix A is-

- (A) A is a zero matrix (B) $A = (-1)I$, where I is a unit matrix
 (C) A^{-1} does not exist (D) $A^2 = I$

Ans. D

Sol. -

Q30. Let $A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{pmatrix}$ and $(10)B = \begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{pmatrix}$. If B is the inverse of matrix A, then α is-

- (A) -2 (B) -1 (C) 2 (D) 5

Ans. D

Sol. $B = A^{-1} \Rightarrow AB = I$

hence product of 2nd row and 3rd column = 0

$$\frac{1}{10} \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$4 + \alpha - 9 = 0 \Rightarrow \alpha = 5 \text{ Ans.}$$

Q31. For a non singular square matrix A of order n, which one of the following statement is true.

- (A) $\text{adj}(KA) = K(\text{adj}A)$ (B) $|\text{adj} A| = |A|$ (C) $(\text{adj} KA) = K^{n-1} \text{adj} A$
 (D) $A.(\text{adj} A) = A^{-1}$

Ans. C

Sol. $|\text{adj} A| = |A|^{n-1}$ (1)

Also $A. \text{adj} A = |A| I_n$

$A \rightarrow kA$

$(KA) \text{adj} (KA) = |KA| I_n$

$(KA) \text{adj} (KA) = K^n |A| I_n$

$A. \text{adj} (KA) = K^{n-1} (\text{adj} A) A$

$\text{adj} (KA) = K^{n-1} (\text{adj} A)$

Q32. If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & a & 1 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & c \\ 4/2 & -3/2 & 1/2 \end{bmatrix}$, then

- (A) $a = 1, c = -1$ (B) $a = 2, c = -\frac{1}{2}$ (C) $a = -1, c = 1$ (D) $a = \frac{1}{2}, c = \frac{1}{2}$

Ans. A

Sol. $AA^{-1} \Rightarrow I \text{ R}_2 \text{ C}_3 = 0$

$$\frac{1}{2} + 2c - \frac{3}{2} = 0$$

$$2c = -2 \Rightarrow c = -1$$

also $\text{R}_2 \text{ C}_2 = 0$

$$-\frac{3}{2} + 3a - \frac{3}{2} = 0$$

$$3a = 3 \Rightarrow a = 1$$

hence $a = 1; c = -1$

Q33. Let P and Q be 3×3 matrices with $P \neq Q$. If $P^3 = Q^3$ and $P^2Q = Q^2P$, then determinant of $(P^2 + Q^2)$ is equal to :

- (A) -1 (B) 2 (C) 1 (D) 0

Ans. D

Sol. $(P^2 + Q^2)P = P^3 + Q^2P$ (1)

$(P^2 + Q^2)Q = P^2Q + Q^3$ (2)

Equation (1) – Equation (2)

$$(P^2 + Q^2)(P - Q) = P^3 - Q^3 + Q^2P - P^2Q$$

$$(P^2 + Q^2)(P - Q) = 0 \quad \because (P \neq Q)$$

$$P^2 + Q^2 = 0$$

$$\text{so } |P^2 + Q^2| = 0$$

Q34. Let A be a square matrix all of whose entries are integers. Then which one of the following is true ?

- (A) If $\det A \neq \pm 1$, then A^{-1} exists and all its entries are non-integers
 (B) If $\det A = \pm 1$, then A^{-1} exists and all its entries are integers
 (C) If $\det A = \pm 1$, then A^{-1} need not exist
 (D) If $\det A = \pm 1$, then A^{-1} exists but all its entries are not necessarily integers

Ans. B

Sol. As $\det(A) = \pm 1$, A^{-1} exists and $A^{-1} = \frac{1}{\det(A)} (\text{adj } A) = \pm (\text{adj } A)$

All entries in $\text{adj}(A)$ are integers.

$\therefore A^{-1}$ has integer entries.

Q35. Let A be a 2×2 matrix with real entries. Let I be the 2×2 identity matrix. Denote by $\text{tr}(A)$, the sum of diagonal entries of A, Assume that $A^2 = I$.

Statement- 1:

If $A \neq I$ and $A \neq -I$, then $\det A = -1$

Statement -2 :

If $A \neq I$ and $A \neq -I$, then $\text{tr}(A) \neq 0$

(A) Statement-1 is true, Statement -2 is true; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is true, Statement -2 is true; Statement-2 is not a correct explanation for Statement-1

(C) Statement-1 is true, Statement -2 is false

(D) Statement-1 is false, Statement-2 is true

Ans. 5eeb5d2b496d5e98d8fa98bc

Sol. $A^2 = I$

$$|A^2| = |I|$$

$$|A^2| = 1$$

$$|A| = 1$$

statement -1 :

If $A \neq I, A \neq -I$

but $|A| = 1$

so this statement is true

statement - 2:

$$\text{Let } A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$|A| = -1 \quad \text{tr}(A) = 0$$

but $A \neq I, A \neq -I$

Q36. The equation $\begin{bmatrix} 1 & 2 & 2 \\ 1 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ has a solution for (x, y, z) besides (0, 0, 0). The value of k equals

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

$$\text{Sol. } \begin{bmatrix} 1 & 2 & 2 \\ 1 & 3 & 4 \\ 3 & 4 & k \end{bmatrix} = 0$$

$$1(3k - 16) - 2(k - 12) + 2(4 - 9) = 0$$

$$3k - 16 - 2k + 24 - 10 = 0$$

$k = 2$ Ans.

Q37. The number of 3×3 matrices A whose entries either 0 or 1 and for which the system

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ has exactly two distinct solutions, is}$$

(A) 0 (B) $2^9 - 1$ (C) 168 (D) 2

Ans. A

Sol. $a_1x + b_1y + c_1z = 1$

$$a_2x + b_2y + c_2z = 0$$

$$a_3x + b_3y + c_3z = 0$$

No three planes can meet at two distinct points. So number of matrices is 0